

AIR



WE LIVE, MOVE, AND BREATHE at the bottom of an immense ocean of air called the atmosphere. Air is an invisible mixture of gases, made up of a teeming mass of millions of tiny gas molecules that move about randomly and at high speed. Without air, the Earth would be a lifeless planet, because the gases air contains are vital to plants and animals.



Carbon dioxide (CO₂)

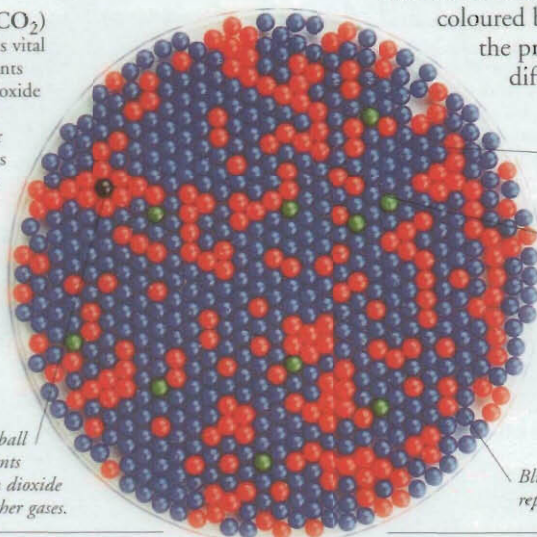
Carbon dioxide is vital for plant life. Plants absorb carbon dioxide from the air and combine it with water gathered by their roots to form sugars, which they use for growth.

Tablets of nitrogen fertilizer

Nitrogen (N₂)

Every living cell contains nitrogen. Plants cannot take nitrogen from the air, so they get it from the soil. Fertilizers contain nitrogen to replenish what plants remove from the soil.

Black ball represents carbon dioxide and other gases.



Red balls represent oxygen.

Green balls represent argon.

Blue balls represent nitrogen.

Fractional distillation

The gases in air have many uses. For example, divers use tanks of oxygen to enable them to breathe underwater, and nitrogen is used in explosives. Gases are extracted from air by a process called fractional distillation. Air is cooled and compressed until it forms a blue liquid. When the liquid expands and warms up, each gas boils off at a different temperature and is collected separately.



Divers with oxygen tanks

Composition of air

Any volume of pure, dry air is 78.09% nitrogen, 20.95% oxygen, 0.93% argon, and 0.03% carbon dioxide and other gases. These coloured balls represent the proportions of the different gases in air.

Candle burns in jar of air.

Flame goes out and water level rises as the oxygen is used up.



Oxygen (O₂)

Burning is a chemical reaction of a substance with oxygen, as this experiment shows. The candle burns in the jar of air until it has used up all the oxygen. Humans and other animals use oxygen from the air to "burn" food inside their bodies and produce energy.

Argon (Ar)

The gas argon is called an "inert" gas because it is so unreactive. Electric light bulbs are often filled with argon. It prevents the bulb's filament from burning up as it would in air, giving the bulb a much longer life.



Air pollution

Air is not naturally "pure" and contains varying amounts of other substances, such as dust, water vapour, bacteria, pollen, and polluting gases. Air pollution from industry and traffic can cause serious health problems in towns and cities, as well as long-term damage to the environment.



Smog

The hazy air pollution that hangs over an urban area is called smog. Sulphurous smog is the result of burning fuels with a high sulphur content, such as coal. Photochemical smog occurs when sunlight causes car exhaust fumes to react together.



Water vapour

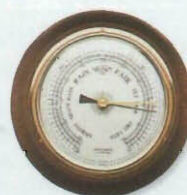
Up to 4 per cent of the volume of air may be water vapour. Warm air can hold more water vapour than cool air. A can of cold drink absorbs heat from the air around it. As the air cools, water vapour condenses out of the air to form droplets on the outside of the can.

Air pressure

Air exerts a force on objects because its moving molecules are constantly colliding with them. Air pressure is a measure of this force. The pressure of the open air is called atmospheric pressure. It is lower at high altitudes, where the air is less dense.

Barometer

A device that measures atmospheric pressure is called a barometer. It can be used to forecast a change in the weather, because air pressure varies slightly from day to day with changes in the air's temperature and humidity.

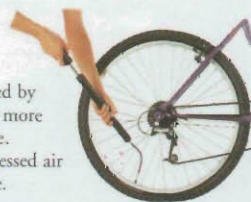


Sucking

When a person sucks on one end of a drinking straw, the lungs reduce the air pressure inside the straw. Atmospheric pressure on the liquid's surface does the rest, pushing down on the liquid, and making it rise up through the straw.

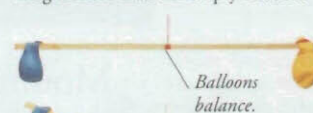
Compressed air

The pressure of air can be increased by compressing it – that is, pumping more and more of it into a limited space. Bicycle tyres are filled with compressed air to give a smooth, comfortable ride.



Weight of air

Air has weight, as this simple experiment proves. Identical empty balloons are attached to both ends of a stick. The balloons balance when the stick is suspended from its middle. Inflating one of the balloons tips the balance, because the balloon full of compressed air weighs more than the empty balloon.



Balance is tipped by inflated balloon.

Inflated balloon contains compressed air.

Joseph Priestley

English scientist and clergyman Joseph Priestley (1733–1804) discovered oxygen in 1774. He also discovered many other gases, including nitrous oxide (laughing gas) and ammonia. Priestley studied carbon dioxide and devised a way to make carbonated (fizzy) water.



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